Serhii Hirus

AI Automation & Backend Engineer · Remote, full-time

serhiihirus@gmail.com · Portfolio · LinkedIn · GitHub

SUMMARY

AI Automation and Backend Engineer with a founder mindset, specializing in Python (FastAPI, Asyncio), LLM orchestration (Claude, OpenAI), and autonomous media/CRM pipelines (n8n, Docker). Proven track record of owning end-to-end systems from raw idea to production, optimizing latency, and shipping reliable solutions that save 15+ hours weekly.

SKILLS

- **Core AI & Automation Stack:** Python (FastAPI, Asyncio, Pydantic), Claude API, OpenAI API, n8n, Docker, PostgreSQL, LangGraph, Whisper, FFmpeg
- **AI agents and LLM pipelines:** Claude API (Anthropic), OpenAI API, LangGraph, tool calling, structured outputs (Pydantic), AST validation, retries and model fallbacks
- **RAG and knowledge bases:** Qdrant, pgvector, LangChain, hybrid search with reranking
- **Automation and integrations:** n8n (self-hosted and cloud), Make, webhooks, REST APIs, HubSpot, Pipedrive, Telegram, Slack
- **Backend and data:** Python (FastAPI, Asyncio, Pydantic), SQL, PostgreSQL, Playwright, JavaScript / Node.js
- **Media and speech:** Whisper, FFmpeg
- **DevOps:** Docker, Linux, Git / GitHub, CI/CD, Google Cloud (GCP)

WORK EXPERIENCE

Independent & Client Contracts — AI Automation & Backend Engineer 2025 - Present

Designing and deploying autonomous AI pipelines, agentic workflows, and CRM integrations for clients and business processes.

Python · FastAPI · Claude API · OpenAI API · n8n · Pydantic · Docker · PostgreSQL

- Engineered async lead processing pipeline (FastAPI, n8n, Apollo API, Whisper): automated audio transcription, BANT metric extraction, and two-way CRM sync, cutting first response time to 30 seconds and saving up to 12 hours weekly per sales rep.
- Architected autonomous multi-platform content pipeline (Whisper, FFmpeg, Claude API, PostgreSQL): automated clipping, transcription, and tone adaptation for 6 platforms, replacing 15+ hours of manual work weekly.
- Built multi-turn domain-specific AI agent with Socratic clarification loops on Claude API and static AST validation, ensuring 100% risk parameter coverage and reducing specification turnaround from hours to < 30 seconds.

Self-employed / FinTech — Quantitative Systems & Python Developer 2022 - 2025

Systematic market analysis and quantitative risk management with transition into algorithmic streaming systems.

Python · Pandas / NumPy · VectorBT · Asyncio · FastAPI

- Designed and executed quantitative risk protocols, drawdown limits, and position sizing models across high-volatility financial sessions with zero unmanaged risk exceptions.
- Engineered streaming market data ingestion and liquidity sweep detectors using Python and Pandas, achieving sub-second alerting latency.
- Prototyped backtesting engines and signal validation logic with VectorBT and AST syntax checks, establishing the software engineering foundation for subsequent AI pipeline automation.

SELECTED PROJECTS

AI Strategy Co-pilot (hirus.systems)

Python · Claude API · AST Validation · VectorBT · Pandas / NumPy · FastAPI

- Multi-turn domain-specific AI agent transforming vague trading hypotheses into strictly verified algorithmic specifications, eliminating hallucinations and unhandled risk boundaries.

Automated Content Creator (demo)

Python · Claude API · Whisper · FFmpeg · n8n · PostgreSQL · Docker · Google Cloud (GCP)

- Built an autonomous pipeline that turns videos and articles into posts and subtitled clips for 6 platforms (LinkedIn, Telegram, X, Threads, Medium, Discord).

AI-copilot for CRM (demo)

Python · FastAPI · n8n · Apollo API · Whisper · HubSpot · Pipedrive · PostgreSQL

- Eliminated manual pre-call lead research and automated CRM post-processing after sales meetings.

EDUCATION

Kherson State Maritime Academy — Bachelor of Science (B.S.) in Marine Engineering 2019 - 2023

CERTIFICATIONS

- Google: Introduction to Generative AI, 2026.
- Coursera: Generative AI with Large Language Models, 2026.

LANGUAGES

English — B1 · Ukrainian — native · Russian — native